

Stakeplot Mobile Project Documentation

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Scope: Finvu account-linking flow and bank transaction APIs, with a project-wide map.

1. Project Map

Area	Path	Main responsibility
Flutter mobile frontend	mobile-frontend/	Screens, app navigation, API route constants, Finvu SDK integration, bank dashboards, transaction history, collections, community, rewards, auth.
Main backend	mobile-backend/mobile-backend/	Core app APIs under /api/v1 : auth, user, community, chat, budget, bill, notifications, bank proxy, email proxy.
Bank service	mobile-backend/bank-service/	Bank/Finvu APIs, transaction automation, bank account storage, categorization, insights, collections, reserve, referral, webhooks.
Email service	mobile-backend/email-service/	Gmail/email token handling, email scraping, statement passwords, pending statement processing, credit card statement extraction.

2. API Base URLs From Flutter

Defined in mobile-frontend/lib/routes/index_route.dart .

Constant	Value pattern	Used for
API.mainBackendUrl	\${Credentials.LIVE_API_TEST or LIVE_API}api/v1	Main backend modules like auth, user, budget, chat, post, split.
API.EmailUrl	\${base}api/v1/email	Email service proxy endpoints.
API.BankApiUrl	\${base}api/v1/bank	Bank service proxy endpoints.
BackendApiEndpoints.finvu	\${API.BankApiUrl}/finvu	Finvu consent and fetch APIs.
BackendApiEndpoints.transactionauto	\${API.BankApiUrl}/transactionauto	Automated bank transaction APIs.
BackendApiEndpoints.transaction	\${API.BankApiUrl}/transaction	Older/manual transaction APIs.

Backend note: the bank service itself mounts routes at /api , while the public mobile app calls them through the main backend proxy at /api/v1/bank/... .

3. Mobile Page Routes

Defined in mobile-frontend/lib/routes/routes.dart .

App route	Screen class	File	Purpose
/discover	DiscoverAccount	finvu_screens/discoverAccount.dart	Shows Finvu-supported banks/FIPs and starts account discovery/linking.
/ShareAccountLogin	ShareAccountLogin	finvu_screens/shareAccountLogin.dart	Entry page for sharing account data with Finvu.
/FetchTransaction	FetchTransaction	finvu_screens/FetchTransaction.dart	Screen shown while fetching linked account transactions.
/OnboardingScreen	OnboardingScreen	onboarding_screens/onboarding_screen.dart	Onboarding destination after linking/fetch flow for new users.
/FinanceDashboard	FinanceDashboard	finance_screen/finance_dashboard/index_finance.dart	Main finance dashboard after bank data exists.
/home	HomePage	Home_Screen/home_screen_state/home_page.dart	Home page, includes bank-linked state and finance widgets.

4. Finvu Frontend API Routes

Defined in `mobile-frontend/lib/routes/route_finvu.dart`.

Flutter route field	Public endpoint	Method	Auth	Purpose
FinvuRoutes.login	/api/v1/bank/finvu/login	POST	Yes	Create a Finvu consent request and return <code>consentHandleId</code> .
FinvuRoutes.fetchData	/api/v1/bank/finvu/fetchData	POST	Yes	Poll consent, create FI request, queue final transaction fetch.
FinvuRoutes.fetchWeekly	/api/v1/bank/finvu/fetchWeekly	POST	Yes	Re-fetch/update bank data for an existing consent/account.
FinvuRoutes.getFipsMetric	/api/v1/bank/finvu/fipsmetric	GET	No backend middleware	Fetch and store latest FIP metrics.
FinvuRoutes.getStatus(id)	/api/v1/bank/finvu/status/:id	GET	No backend middleware	Check consent status using stored handle record id.
FinvuRoutes.getFipDetails	/api/v1/bank/finvu/fip-details	POST	Yes	Get latest metrics for selected FIP ids.
FinvuRoutes.addFinvuData	/api/v1/bank/finvu/add	N/A	N/A	Frontend constant exists, but no matching backend route was found.

5. Finvu Backend Routes

Defined in `mobile-backend/bank-service/src/routes/finvu-routes.ts`.

Met hod	Public mobile route	Controller	Request body / params	Response notes
PO ST	/api/v1/bank/finvu/login	loginAndGetHandleId	custId , number	Returns consentHandleId and metric . Creates Finvu consent via /ConsentRequestPlus .
PO ST	/api/v1/bank/finvu/fetchData	fetchTransactions	handleId , custId	Polls consent, creates FI request, writes Redis/Mongo fetch state, returns sessionId .
PO ST	/api/v1/bank/finvu/fetchWeekly	fetchTransactionsWeekly	custId , userId , consentId , handleId , FROM , optional metadata	Starts update FI request for existing consent and returns fetch progress data.
PO ST	/api/v1/bank/finvu/fip-details	getFipsDetails	{ "fipIds": ["..."] }	Returns latest metric rows for selected FIPs.
GET	/api/v1/bank/finvu/fipsmetric	getFipsLatestMetricsAll	None	Calls Finvu /fips/latest-metrics-all , drops and reinserts metric collection.
GET	/api/v1/bank/finvu/status/:id	getStatus	id = ConsentHandleId Mongo id	Loads stored handle and checks Finvu consent status.

6. Finvu Controller Flow

Initial consent flow

1. Flutter calls `getConsentHandleId(context)` in `finvu_screens/integration.dart` .
2. The function posts to `FinvuRoutes.login` with `custId` and `number` .
3. Backend `loginAndGetHandleId` generates a Finvu token, calls `ConsentRequestPlus` , stores the handle, fetches FIP metrics, and returns `consentHandleId` .

```
{
  "custId": "<mobile>@finvu",
  "number": "<mobile>"
}
```

Finvu SDK login and account linking

1. Flutter initializes `finvuManager` in `initFinvuManager` .
2. Flutter calls `finvuManager.loginWithUsernameOrMobileNumberAndConsentHandle` .
3. OTP reference is stored in `otpReference` .
4. `verify(otp, context)` calls `finvuManager.verifyLoginOtp` .
5. After OTP success, Flutter fetches linked accounts and navigates to `DiscoverAccount` .

Transaction fetch flow

1. Flutter calls `FetchTransactionFromFinvuApi(context)` .
2. It marks the user fetch state using `UserRoutes.updateFetchStatus` .
3. It posts to `FinvuRoutes.fetchData` .

4. Backend polls consent status, gets consent details, initiates FI request, stores `finvu:<sessionId>` in Redis, stores `Finvu` in Mongo, queues the fetch in non-production, and returns `sessionId`.

```
{
  "token": "",
  "handleId": "<consentHandleId>",
  "custId": "<mobile>@finvu"
}
```

Final data fetch / webhook path

Trigger	Location	Purpose
Finvu webhook	POST <code>/api/FI/Notification</code> to <code>webHook</code>	Production data-ready notification path.
Bull queue fallback	<code>services/bull-queue-service/finvu-fetch-queue.ts</code>	Non-production delayed final fetch path.
Manual queue trigger	<code>/api/finvu-queue/trigger</code>	Staging test route.

7. Bank Transaction Frontend Routes

Defined in `mobile-frontend/lib/routes/route_transactions.dart`. Base: `/api/v1/bank/transactionauto`.

Flutter route field	Endpoint	Backend match
<code>getAutoPays</code>	GET <code>/autopays</code>	Yes
<code>createRecurringPaymentFromTransaction</code>	POST <code>/recurring-payments/from-transaction/:transactionId</code>	Yes
<code>categorizeGroupedTransaction</code>	POST <code>/grouped/:groupId/categorize</code>	Yes
<code>verifyPendingTransaction</code>	POST <code>/verify-pending-transaction/:transactionId/:isCorrect</code>	Yes
<code>updateTransaction</code>	PATCH <code>/updateTransaction/:transactionId</code>	Yes
<code>getSearchedTransactions</code>	GET <code>/getTransactions/:page</code>	Yes
<code>getTransactionsOfUser</code>	GET <code>/getTransactionsOfUser</code>	Yes
<code>categorizeTransactions</code>	GET <code>/categorize</code>	Yes
<code>getGroupedTransactions</code>	GET <code>/get-grouped-transactions</code>	Yes
<code>getPendingForReviewTransactions</code>	GET <code>/pending-for-review-transactions</code>	Yes
<code>getDayWiseTransactionsSummary</code>	GET <code>/get-day-wise-transactions</code>	Yes
<code>getTransactionsByDate</code>	GET <code>/get-day-wise-transactions/:date</code>	Yes
<code>getRecurringPayments</code>	GET <code>/get-recurring-payments/:isActive</code>	Yes
<code>updateRecurringPayment</code>	PATCH <code>/recurring-payments/:id</code>	Yes
<code>deleteRecurringPayment</code>	DELETE <code>/recurring-payments/:id</code>	Yes
<code>getLoanCalculation</code>	POST <code>/get-loan-calculation</code>	Yes

Flutter route field	Endpoint	Backend match
getAllCustomTransactions	GET /getAllCustomTransactions/:accountId/:type/:value	Yes
getWholeTransactionsGraph	GET /getWholeTransactionsGraph/:type/:value	Yes
getBanksLinkedAndAccounts	GET /get-banks-linked	Yes
getQuickCheck	GET /get-banksdebitcredit	Yes
getMonthlyTransactionsHistory	GET /get-monthly-transactions-history/:accountId/:type/:page	Yes
getPreviousTransactions	GET /get-previous-transactions/:date/:accountId	Yes
getTopThreeTransactionsOfWeek	GET /top-three-transactions-of-week	Yes
getIncomeAndCategorySpent	GET /get-income-average-monthly-category-expenses	Yes
deleteBankAccount	DELETE /:bankId/:accountId	Yes
deleteTransactions	POST /delete	Yes

Known mismatches: get-hide-transactions , getTransactionsForAccount , month/week routes, heads-up/money-map routes, and FinvuRoutes.addFinvuData are present in frontend constants but not in the referenced backend route files.

8. Bank Transaction Backend Routes

Defined in `mobile-backend/bank-service/src/routes/transactionsAuto/transactions.ts` . All routes use auth unless marked as unauthenticated.

Meth od	Public mobile endpoint	Controller	Purpose
GET	/api/v1/bank/transactionauto/autopays	getAutoPays	Fetch detected auto-pay/recurring debit groups.
POST	/api/v1/bank/transactionauto/recurring-payments/from-transaction/:transactionId	createRecurringPaymentFromTransactionController	Create recurring payment record from a transaction.
POST	/api/v1/bank/transactionauto/grouped/:groupId/categorize	categorizeGroupedTransaction	Categorize a grouped set of transactions.
POST	/api/v1/bank/transactionauto/verify-pending-transaction/:transactionId/:isCorrect	verifyPendingTransaction	Mark auto/pending transaction prediction as correct or incorrect.
PATCH	/api/v1/bank/transactionauto/updateTransaction/:transactionId	updateTransaction	Update category, subcategory, hidden/excluded fields, etc.
GET	/api/v1/bank/transactionauto/getTransactions/:page	getSearchedTransactions	Fetch paginated searched/filtered transactions.
GET	/api/v1/bank/transactionauto/getTransactionsOfUser	getAllTransactionsOfUser	Fetch all transactions for current user.

Meth od	Public mobile endpoint	Controller	Purpose
GET/ POST	/api/v1/bank/transactionauto/dummy-categorize-preview	dummyCategorizationPreview	Unauthenticated categorization preview.
GET	/api/v1/bank/transactionauto/categorize	categorizeTransactions	Run or fetch categorization logic for transactions.
POST	/api/v1/bank/transactionauto/create	createTransaction	Create a transaction manually/custom.
GET	/api/v1/bank/transactionauto/top-five-categories	getTopFiveCategories	Top spend categories.
GET	/api/v1/bank/transactionauto/category-wise-spending	getCategoryWiseSpending	Category-wise spending summary.
GET	/api/v1/bank/transactionauto/budget-transactions	getBudgetTransactions	Transactions related to budgets.
GET	/api/v1/bank/transactionauto/get-budget-spends	getBudgetSpends	Budget spending totals.
PUT	/api/v1/bank/transactionauto/:transactionId	updateTransaction	Alternate transaction update route.
GET	/api/v1/bank/transactionauto/getUserMonthlySpending	getUserMonthlySpending	Monthly spending summary.
GET	/api/v1/bank/transactionauto/get-grouped-transactions	getGroupedTransactions	Group similar transactions.
GET	/api/v1/bank/transactionauto/pending-for-review-transactions	getPendingForReviewTransactions	Transactions awaiting user review.
GET	/api/v1/bank/transactionauto/get-day-wise-transactions	getDayWiseTransactionsSummary	Day-wise summary for calendar/dashboard.
GET	/api/v1/bank/transactionauto/get-day-wise-transactions/:date	getTransactionsByDate	Transactions for a selected date.
GET	/api/v1/bank/transactionauto/get-recurring-payments/:isActive	getRecurringPayments	Active/inactive recurring payment list.
PATCH	/api/v1/bank/transactionauto/recurring-payments/:id	updateRecurringPayment	Update recurring payment.
DELETE	/api/v1/bank/transactionauto/recurring-payments/:id	deleteRecurringPayment	Delete recurring payment.
POST	/api/v1/bank/transactionauto/get-loan-calculation	getLoanCalculation	Loan calculation based on income/spend.
GET	/api/v1/bank/transactionauto/getAllCustomTransactions/:accountId/:type/:value	getAllCustomTransactions	Account-specific graph/custom-date transactions.
GET	/api/v1/bank/transactionauto/getWholeTransactionsGraph/:type/:value	getWholeTransactionsGraph	Overall graph data.

Meth od	Public mobile endpoint	Controller	Purpose
GET	/api/v1/bank/transactionauto/user-details	getUser	Current user detail from bank context.
GET	/api/v1/bank/transactionauto/get-banks-linked	getBanksLinkedAndAccounts	Linked banks and account details.
GET	/api/v1/bank/transactionauto/get-banksdebitcredit	getBankBalanceAndDebitSummary	Bank balance/debit/credit summary.
GET	/api/v1/bank/transactionauto/get-monthly-transactions-history/:accountId/:type/:page	getMonthlyTransactionsHistory	Paginated monthly/yearly/custom transaction history.
GET	/api/v1/bank/transactionauto/get-previous-transactions/:date/:accountId	getPreviousTransactions	Previous transactions from selected date/account.
GET	/api/v1/bank/transactionauto/top-three-transactions-of-week	getTopThreeTransactionsOfWeek	Top three weekly transactions.
GET	/api/v1/bank/transactionauto/get-income-average-monthly-category-expenses	getIncomeAndCategorySpent	Income and average monthly category expenses.
DELETE	/api/v1/bank/transactionauto/:bankId/:accountId	deleteBankAccount	Delete one bank account and related data.
POST	/api/v1/bank/transactionauto/delete	deleteTransactions	Bulk delete transactions.

9. Other Bank Routes Used By Transaction Code

Manual/legacy transaction routes

Method	Endpoint	Controller	Purpose
POST	/api/v1/bank/transaction/add	enterTransaction	Add manual transaction.
POST	/api/v1/bank/transaction/updateGroupTransactions/:id	updateGroupTransaction	Update group transaction amount.
GET	/api/v1/bank/transaction/all	getAllTransactions	Fetch old transaction list.
DELETE	/api/v1/bank/transaction/:id	deleteSpecificTransaction	Delete a specific old transaction.

Custom category routes

Method	Endpoint	Controller	Purpose
POST	/api/v1/bank/custom/custom-category	addCustomCategory	Add user custom category.
GET	/api/v1/bank/custom/custom-category	getCustomCategories	Fetch user custom categories.

10. Screen / Repository To API Mapping

Frontend file	Functions/classes	API routes used
finvu_screens/integration.dart	getConsentHandleId , FetchTransactionFromFinvuApi , login , verify	FinvuRoutes.login , FinvuRoutes.fetchData , UserRoutes.updateFetchStatus , Finvu SDK login/OTP APIs.
repository/bankinfo.dart	getBankAccounts	BankTransactionRoutes.getBanksLinkedAndAccounts .
repository/bankinfo.dart	getWeeklyfetchData	FinvuRoutes.fetchWeekly , UserRoutes.updateFetchStatus .
repository/bankinfo.dart	getFipAccountInfo	FinvuRoutes.getFipDetails .
repository/transactions_repository.dart	transaction history and filters	getSearchedTransactions , getMonthlyTransactionsHistory , getDayWiseTransactionsSummary , getTransactionsByDate .
repository/transactions_repository.dart	update/tag/hide/exclude	updateTransaction , categorizeGroupedTransaction , verifyPendingTransaction , deleteTransactions .
repository/autopay_repository.dart	auto-pay CRUD	getAutoPays , createRecurringPaymentFromTransaction , updateRecurringPayment , deleteRecurringPayment .
repository/delete_banks_users.dart	bank deletion	deleteBankAccount .
finance_repository.dart , expanded_finance_repository.dart	graphs/custom finance views	getAllCustomTransactions .
finance_screen/Calculators/loan_calculator.dart	loan/income calculation	getIncomeAndCategorySpent , getLoanCalculation .

11. Important Data Models / Storage

Model/storage	Used for
ConsentHandleId	Stores custId , consent handle id, user id, expiry for Finvu consent status lookup.
Finvu	Stores FI fetch sessions and metadata for async processing.
FipsMetric	Stores latest FIP metrics fetched from Finvu.
Bank , Account , Profile , Summary , Transaction	Persist linked bank, account, profile, balance summary, and transaction records.
Redis key auth_token	Finvu auth token used by fetch/status calls.
Redis key finvu:<sessionId>	Temporary FI request metadata for queue/webhook final fetch.
Redis pubsub channel bank_events	Publishes websocket events for fetch success/failure.

12. Environment Variables Used By Finvu

Variable	Purpose
<code>FINVU_URL</code>	Base Finvu API URL.
<code>FINVU_RID</code>	Finvu request header <code>rid</code> .
<code>FINVU_TS</code>	Finvu request header <code>ts</code> .
<code>FINVU_CHANNEL_ID</code>	Finvu request header <code>channelId</code> .
<code>FINVU_CONSENT_DESCRIPTION</code>	Consent request description.
<code>FINVU_TEMPLATE_NAME</code>	Finvu consent template.
<code>FINVU_USER_SESSION_ID</code>	User session id included in consent request.
<code>NODE_ENV</code>	Controls whether fallback Bull queue is triggered for non-production.

13. Known Gaps / Follow-up Items

Item	Why it matters
<code>FinvuRoutes.addFinvuData</code> has no backend route.	Remove the unused frontend constant or add the missing backend endpoint.
<code>fipsmetric</code> and <code>status/:id</code> are unauthenticated in <code>finvu-routes.ts</code> .	Consider adding auth if these expose sensitive operational or user-specific data.
Several frontend <code>BankTransactionRoutes</code> constants do not match current backend routes.	Causes runtime 404s if screens call them.
<code>customCategory</code> frontend base points to <code>API.mainBackendUrl</code> , but backend route is in bank service under <code>/custom</code> .	Verify the main backend proxy supports <code>/api/v1/custom</code> ; otherwise change to <code>API.BankApiUrl</code> .
Public mobile path goes through <code>/api/v1/bank</code> , while bank service internal mount is <code>/api</code> .	Keep docs and route constants explicit to avoid confusing service-local paths with public app paths.

14. Quick End-to-End Sequence

1. User opens Finvu sharing flow from `ShareAccountLogin` .
2. User enters mobile number in `MobileNumber` .
3. Flutter calls backend `POST /api/v1/bank/finvu/login` .
4. Backend creates Finvu consent and returns `consentHandleId` .
5. Flutter uses Finvu SDK to login with mobile number and handle id.
6. User verifies OTP in `FinvuVerifyOtpScreen` .
7. Flutter discovers and links accounts through Finvu SDK screens.
8. User confirms access in `Access` .
9. Flutter calls `POST /api/v1/bank/finvu/fetchData` .
10. Backend creates FI request and stores `sessionId` .
11. Webhook or Bull queue fetches final FI data.
12. Bank service stores `bank/account/profile/summary/transactions`.

13. Flutter finance/dashboard/history screens call `transactionauto` endpoints to show the data.